

ILYAS ARYA

Indonesia | bashirahilyas@gmail.com | +62 811-1288-757 | linkedin.com/in/ilyas.arya | github.com/milyasbpa

Senior Frontend Engineer focused on web application architecture, with experience owning high-traffic booking funnels and secure banking portals, decoupling multi-step state models and server contracts, and driving cross-team delivery.

SELECTED IMPACT

- Owned the frontend architecture of Malaysia Airlines' multi-step booking funnel across 6 checkout modules, decoupling URL state, server data, and persisted session drafts to eliminate cross-step state corruption and ensure zero data loss during browser navigation.
- Drove web performance optimization for AI-powered image annotation applications by offloading heavy image processing workloads to Web Workers, achieving 0ms main-thread UI blocking and smooth 60 FPS interactions during live analysis.
- Created and governed an enterprise React & TypeScript UI component library across core banking portals at Bank Rakyat Indonesia, standardizing design tokens, accessible component APIs, and testing gates adopted across multiple engineering squads.

PROFESSIONAL EXPERIENCE

Trinity Wizards, Senior Frontend Developer, Jan 2024 – Present

Technologies: Next.js, TypeScript, Redux, WebSockets, Amadeus GDS, MOP SDK

- Architected a multi-step booking funnel for Malaysia Airlines integrating Amadeus APIs across cart, booking, and extra services modules, decoupling URL parameters, server state, and persisted session drafts across 6 checkout modules to eliminate cross-step state contamination.
- Engineered real-time seat selection UI using WebSockets and component virtualization, handling concurrent seat-locking mechanisms without main-thread UI degradation or double-booking race conditions.
- Integrated payment processing using the MOP (Means of Payment) SDK, implementing automatic session recovery and error boundaries to secure multi-currency transactions during checkout.

Alvisual, Senior Frontend Developer, Jan 2022 – Jan 2024

Technologies: React.js, TypeScript, PWA, Web Workers, IndexedDB, Canvas API

- Engineered a cross-platform PWA for AI-powered face and scalp analysis deployed across retail outlet tablets in Singapore (iPad/Windows Surface), offloading 10,000+ image coordinate calculations to Web Workers to enable instant annotation toggling and smooth 60 FPS stylus editing while bypassing Safari WebKit canvas memory limits.
- Architected PWA deployment architecture across Kubernetes clusters, implementing versioned asset caching and Service Worker lifecycle update strategies to eliminate stale cache issues and ChunkLoadErrors during live retail tablet updates.
- Designed an offline-first PWA data synchronization architecture, queueing 10,000+ scan coordinate payloads in IndexedDB during retail Wi-Fi dropouts to prevent data loss and ensure uninterrupted tablet consultations.

Bank Rakyat Indonesia, Technical Lead, Jan 2021 – Dec 2021

Technologies: React.js, TypeScript, Redux, Design Tokens, SonarQube, Map APIs

- Created and governed an enterprise React & TypeScript UI component library across AI banking portals at Bank Rakyat Indonesia, standardizing design tokens, accessible component APIs, and usage documentation adopted across multiple engineering squads.
- Architected data-heavy dashboard UIs and interactive geolocation maps for BRI's AI Department, rendering daily AI-targeted customer leads and location routing for field officers (Mantri) with zero UI latency and resilient state hydration.
- Led a team of 5+ frontend engineers in the AI department, establishing a PR review assignment process and review checklist to eliminate PR bottlenecks and enforce SonarQube quality gates prior to release.

DDTC, Lead Frontend Developer, Jun 2020 – Dec 2020

Technologies: Nuxt.js, Vue.js, TypeScript, Vuex, SSR, Technical SEO

- Led frontend development for 5+ tax management applications using Nuxt.js and TypeScript, standardizing SSR architecture and API data hydration across client portals.
- Optimized technical SEO and rendering performance for public tax web portals, integrating Google Analytics event tracking to drive marketing insights and search visibility.

Telkom Indonesia, Frontend Engineer, Aug 2019 – Jun 2020

Technologies: React.js, TypeScript, Redux, React Router, REST APIs

- Developed responsive customer-facing web portals using React.js and TypeScript, integrating RESTful APIs to display account usage metrics and service dashboards.
- Implemented client-side routing, Redux state management patterns, and HTTP session handling across core telecommunication user modules.

TECHNICAL SKILLS

- **Frontend Core:** JavaScript (ES6+), TypeScript, React.js, Next.js, Nuxt.js, Redux, Zustand, HTML5, CSS3
- **Architecture & Performance:** State Modeling, Component APIs, Web Workers (Multithreading), PWA (Offline-First), WebSockets, Code Splitting, Bundle Optimization
- **Quality & Testing:** Cypress, Jest, Playwright, SonarQube Quality Gates, Unit & Integration Testing
- **Platform & DevOps:** Docker, Kubernetes, Jenkins CI/CD, GCP, AWS, Webpack, Vite
- **API & Governance:** RESTful APIs, Amadeus GDS, MOP SDK, PR Review Checklists, Design System Governance

EDUCATION

Bandung Institute of Technology (ITB), Bachelor of Geophysical Engineering, Graduated: April 2020